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To the PhD Selection Committee
EDT Program - Université Côte d'Azur
Attn: Dr. Gérald Rocher

May 1, 2026

Subject: Application for PhD: Distributed Abductive Reasoning for Self-Explaining Digital Twins (Rank 3 - PC1)

Dear Dr. Rocher,

As a State Engineer in Artificial Intelligence from ESI-SBA, I am writing to express my enthusiastic application for the PhD position in "Distributed Abductive Reasoning for Self-Explaining Digital Twins" within the Engineering Digital Twin (EDT) national program. This application is motivated by the synergy between my background in edge-deployed hybrid models and the objectives of your research group.

Currently, I am completing my thesis at Sonatrach, where I have developed a **Physics-Informed Digital Twin** for gas turbine monitoring. A core component of my research is the deployment of generative surrogate models (Mamba-SSM/xLSTM) on **resource-constrained edge devices**, achieving latencies as low as 0.04ms. This experience has provided me with practical insights into the "reality gap" that occurs when hybrid models encounter unforeseen sensor anomalies or system degradations in the field.

I am particularly fascinated by your proposal to use **abductive reasoning** to interpret these discrepancies. My technical background in **Distributed AI** and real-time monitoring aligns perfectly with your vision of a hierarchical explanation architecture spanning edge, fog, and cloud layers. I am eager to contribute to the development of self-explaining digital twins that can transform raw prediction errors into human-interpretable knowledge, thereby increasing trust in automated industrial decision-making.

My background in distributed systems and my experience with production-grade PyTorch and C++ implementations provide me with the necessary tools to contribute to the **Artemis platform** and the broader EDT ecosystem. I am a highly autonomous researcher, comfortable bridging the gap between symbolic reasoning and industrial physical simulations.

Thank you for your time and consideration. I look forward to the possibility of discussing how my experience in edge-deployed AI and hybrid modeling can contribute to the success of your research group in Nice.

Sincerely,

Ismail AISSAOUI